Surface Generation from Point Clouds

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The neural network reconstructor produces cleaner surfaces because it learns a globally consistent signed distance function by aggregating information across all training samples, rather than relying on a single nearest neighbor as the naive method does. Its skip connection preserves fine geometric detail, while dropout and weight normalization promote smooth generalization. The clamped L1 loss focuses training on the critical near-surface region, yielding more accurate and artifact-free mesh extraction — particularly noticeable with sparse point clouds.

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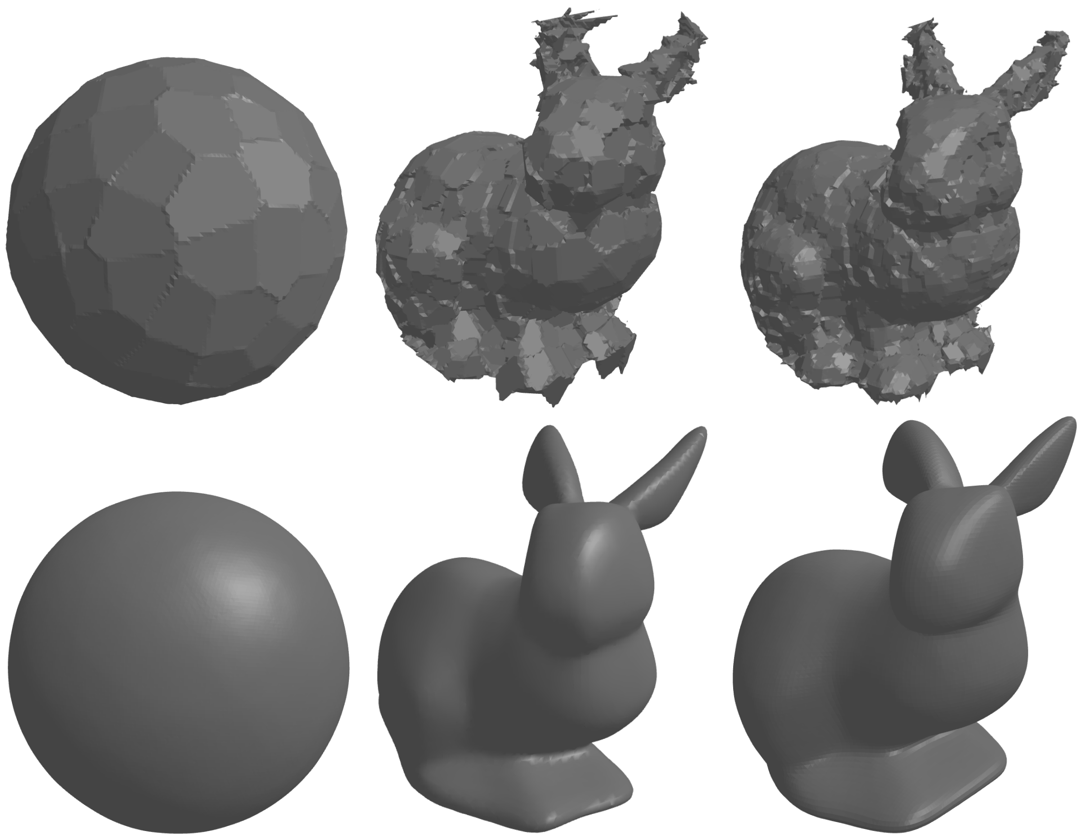

Figure 1: (top row) Naive reconstructor, in the middle is the 500pts cases **(buttom row)** NN reconstructor